

PROJECT PROPOSAL (Discussion Draft)

Project Title : Explainable AI-Based Personal Financial Copilot for Predictive Budgeting, Goal Forecasting, and Financial Risk Analysis

Introduction

Managing personal finances is an important part of everyday life, but many people still find it difficult to track expenses, plan savings, and achieve long-term financial goals. Although several budgeting and expense-tracking applications are available today, most of them only record past transactions and generate simple reports. They rarely help users understand future financial situations or explain the reasoning behind their recommendations.

With the advancement of Artificial Intelligence (AI), Machine Learning (ML), Explainable AI (XAI), and Large Language Models (LLMs), there is an opportunity to transform traditional budgeting applications into intelligent financial assistants. Instead of simply tracking expenses, these systems can predict future spending, identify financial risks, and provide personalized financial guidance while explaining how every recommendation is generated.

This project proposes an **Explainable AI-Based Personal Financial Copilot** that assists users in managing their finances through predictive budgeting, goal forecasting, financial risk analysis, and transparent AI-driven recommendations.

Problem Statement

Most existing personal finance applications focus on expense tracking and budget monitoring. While these features are useful, they do not provide proactive financial planning or personalized decision support. Users often receive predictions without understanding why they were generated, reducing trust in AI-based financial systems.

The major limitations identified in existing systems are:

- Basic expense tracking with limited predictive capabilities.
- Lack of transparent AI recommendations.
- Weak long-term financial goal forecasting.
- Limited financial risk analysis.
- Minimal personalized financial guidance.

These limitations highlight the need for a more intelligent and explainable financial management system.

Proposed Solution

The proposed system is an AI-powered Personal Financial Copilot that moves beyond traditional budgeting applications by helping users make informed financial decisions.

The system will provide:

- Predictive budgeting based on historical spending patterns.
- Goal achievement forecasting for savings and future financial plans.
- Financial risk analysis to identify overspending and cash-flow issues.
- Explainable AI (XAI) that clearly explains every prediction and recommendation.
- A conversational AI assistant that allows users to interact naturally with their financial information and receive personalized guidance.

The objective is not only to predict financial outcomes but also to improve user trust and financial literacy through transparent and understandable AI.

Objectives

The proposed project aims to:

- Develop an intelligent personal financial management system.
- Predict future budgets using machine learning techniques.
- Forecast financial goal achievement based on user spending behavior.
- Analyze financial risks such as overspending and insufficient savings.
- Integrate Explainable AI to improve transparency and user trust.
- Provide personalized financial recommendations through a conversational AI interface.

Scope of the Project

The project focuses on developing a user-friendly financial assistant capable of:

- Expense and income management.
- Predictive budgeting.
- Goal forecasting.
- Financial risk analysis.
- Explainable AI-based recommendations.
- Personalized financial guidance.
- Interactive conversational support.

The proposed system is intended for students, salaried employees, freelancers, and families who wish to improve their financial planning and decision-making.

Expected Contribution

Compared to existing personal finance applications, the proposed project aims to provide a more comprehensive and intelligent solution by combining multiple AI technologies into a single platform.

The expected contributions include:

- An Explainable AI framework for transparent financial recommendations.
- Integration of predictive budgeting, goal forecasting, and financial risk analysis.
- Improved financial decision support through personalized recommendations.
- Increased user confidence by explaining AI-generated predictions.
- A conversational AI assistant for natural and interactive financial guidance.

Future Scope

The system can be further enhanced by integrating technologies such as:

- Open Banking APIs.
- Voice-based financial assistants.
- Blockchain for secure financial records.
- Federated learning for privacy-preserving AI.
- Real-time credit score prediction.
- Multi-modal AI using text, voice, receipts, and transaction history.

These improvements could eventually transform the system into a fully autonomous personal financial assistant.

Reason for Choosing This Project

We chose this project because personal financial management is a real-world problem faced by almost everyone. Although budgeting applications are widely available, they often fail to provide meaningful financial guidance or explain the reasoning behind their recommendations.

This project combines Artificial Intelligence, Explainable AI, and Financial Technology to build a practical solution that helps users make smarter financial decisions while increasing trust in AI systems. It also provides an opportunity to strengthen our knowledge of Machine Learning, Explainable AI, Predictive Analytics, and FinTech by working on a project with real-world applications and social impact.

Potential Base Papers for Discussion

After conducting an initial literature survey, the following papers have been shortlisted for discussion with our project mentor before selecting the final base paper:

1. **Smart Expense Tracker with Spending Pattern Prediction (2026)**
2. **arXiv:2406.11903**
3. **IEEE Paper – 11506666**
4. **IEEE Paper – 11294369**
5. **Springer – 10.1007/s42521-025-00146-3**
6. **SSRN – 6615358**

These papers were selected because they focus on AI-based personal finance management, predictive analytics, Explainable AI, and intelligent financial assistance. Based on our mentor's suggestions, we will finalize the most suitable base paper, refine the project scope, and identify the research gap that our project will address.